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卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0503 30-B003701S-T0503

图纸文件目录（首页）

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拉合尔±660kV换流站 工程 PROJECT　施工图 阶段 DESIGN STAGE
±660kV Lahore Converter Station　Detail Design

结构 专业 Structure SPECIALITY　第 05 卷 VOLUME　第 03 册 BOOK

卷册名称 Volume name：换流变事故油池图 Converter accident oil pool

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图纸 DRAWING NOS. 4 张	打印件 PRINTED COPIES NOS. 0 本

序号 No.	图号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
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注:

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地下构筑物结构设计说明

1、本构筑物设计依据：水工提资及地质勘测报告、国家规范规程。
2、场地±0.000相当于绝对标高193.40(EGM2008高程系统)。如无特别说明，本工程标高均采用相对标高。
3、采用巴基斯坦UTM-43N平面坐标系，相对平面坐标以米(m)为单位，尺寸以毫米(mm)为单位。
4、根据《巴基斯坦建筑规范》(2007)，换流站属于ZONE 2A区，场地土类型为SD类。根据巴基斯坦地面峰值加速度图，地面峰值加速度为0.14g，相当于国内地震基本烈度为7度。
5、本构筑物安全等级二级，设计使用年限50年。地下工程防水等级二级。地基基础设计等级丙级。
6、设计荷载：地面活荷载10kN/m²。
7、基础底应置于良好土层上，如遇不良地基请及时通知设计院。施工期间应注意采取措施防止构筑物上浮。
8、混凝土标号：fc'=28MPa，技术要求：(1)水胶比不大于0.50，(2)氯离子含量不大于0.1%，(3)水泥型号应满足ACI-318-14表19.3.2.1中暴露类别S1对应水泥(Cementitious materials-Types)的要求，(4)抗渗等级0.8MPa，(5)掺矿物掺合料。
混凝土垫层fc'=21MPa。钢筋Φ代表Grade 60 钢，技术要求：应满足ASTM A706, Grade 60[420]变形钢筋或满足ACI 318-14第20.2.2.5节要求的ASTM A615, Grade 60[420]变形钢筋。
钢筋的抗拉强度实测值与屈服强度实测值的比值不应小于1.25；钢筋的屈服强度实测值与屈服强度标准值的比值不应大于1.3，且钢筋在最大拉力下的总伸长率实测值不应小于9%。
钢筋Φ代表Grade 40，钢材Q235B，电焊条E43XX，焊缝高度Hf=h_{min}。
所有外露铁件均热镀锌防腐，焊伤处防锈漆一道，银粉漆二道。
9、防水混凝土应连续浇筑，宜少留施工缝。当留施工缝时不应留在剪力与弯矩最大处或底板与侧板的交接处，应留在高出底板表面不小于300mm的侧墙上。侧墙有预留空洞时，施工缝距孔洞边缘不应小于300mm。
10、受力钢筋的混凝土保护层厚度
顶板上、下均为35mm；底板上为50mm，下为75mm；钢筋混凝土梁、柱50mm。
池壁50mm。
11、池壁四周1m范围填土应采用透水性好的土壤。
应对称分层回填，层厚不大于300mm，压实系数不小于0.94。
12、本构筑物施工验收时应遵守现行施工验收规范。
13、本工程结构施工时应与建筑、水、暖、电等其他工种图纸配合施工。
14、其它未尽事宜请严格按照有关的规程规范、标准图集等的相关内容执行。
15、本工程初设设计尚未审定，未通过NTDC审核，仍存在修改的可能，施工前请联系业主与设计确认。
16、池壁及板顶钢筋在管孔处尽量绕开，不应截断。当开洞直径不大于300mm时，可将受力钢筋绕过洞边，不需切断并可不设洞口的补强钢筋；当开洞直径大于300mm，且洞口周边无集中荷载时，应在洞口每侧配置补强钢筋。
17、受拉钢筋的锚固长度 L_a(mm)：

钢筋种类	抗震等级	钢筋直径	混凝土强度fc 21.0	28.0
		10	53d	52d
		13	53d	45d
		16	53d	45d
		19	53d	45d
		22	52d	45d
		25	52d	45d

注
1.任何情况下，纵向受拉钢筋的锚固长度不应小于250mm。
2.当钢筋在混凝土施工过程中易受扰动时，表中锚固长度应乘以修正系数1.1。
3.纵向受拉钢筋末端采用弯钩或机械锚固措施时，包括弯钩或锚固端头在内的锚固长度(投影长度)可取表中数字的0.6倍。
4.纵向受压钢筋的最小锚固长度不应小于表中数字的0.7倍。

18、纵向受压钢筋的最小搭接长度=0.7L_a，且≥200mm。

纵向受拉钢筋的搭接长度 L_l　　$L_l=L_aX$ 1.2 (≤25%) / 1.4 (50%) / 1.6 (100%)
括号内数字为纵向钢筋搭接接头面积百分率

Specification of structural design for underground structures

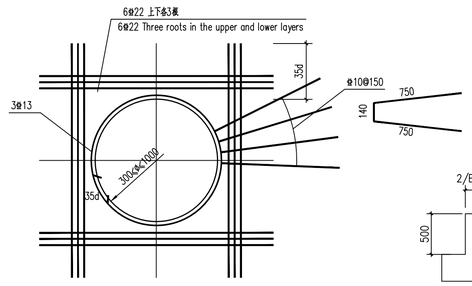
1. Design basis of this structure: Hydraulic capital and geological survey report, the national code of practice.
2. The site level of ±0.000 is equal to the absolute elevation of193.40(geodestic datum system EGM2008) If not specified,the elevations of the project are adopt to the relative elevation.
3. Using Pakistan UTM-43N plane coordinate system,the relative plane coordinates are in meters (m) and the dimensions are in millimeters (mm).
4. According to the "Pakistan Building Code" (2007), the converter station belongs to zone ZONE 2A and the site soil type is SD.According to Pakistan's ground peak acceleration map, the ground peak acceleration is 0.14g, equivalent to the basic earthquake intensity of 7 degrees.
5. The structure safety level: second-class, Design life : 50 years, Underground waterproof rating: second-class, Foundation design level : C.
6. Design load: Floor live loads: 10kN/m².
7. Basal bottom should be placed in good soil, in case of poor foundation, please notify the design institute. Measures should be taken to prevent the building from floating during construction.
8. Strength grading of concrete: fc'=28MPa, technical requirements: (1)the water binder ratio is not greater than 0.50, (2)the content of chloride ion is not more than 0.1%, (3)the cement shall meet the requirements of the exposure category S1 in19.3.2.1 in Table ACI-318-14, corresponding to the cement (Cementitious, materials-Types), (4)the impermeability grade is 0.8MPa, (5)the mineral admixture is added.
The strength of concrete used in cushion is fc'=21Mpa.
Reinforcement Φ: ASTM A615 Grade 60, fy = 420MPa., technical requirements: (1)It shall meet the requirements of ASTM A706, Grade 60[420] deformed steel bars or meeting the requirements of section ACI 20.2.2.5 318-14.The ratio between measured tensile strength and measured value of yield strength should not be less than 1.25;The ratio between the measured yield strength and the standard value of yield strength should not exceed 1.3; The measured value of the total elongation at the maximum tensile force should not be less than 9%.
Reinforcement Φ: ASTM A615 Grade 40, Steel: Q235B, welding rod: E43XX, weld height: Hf=h_{min}.
All exposed iron parts are hot-dip galvanized anti-corrosion,welding wound apply a antirust paint and two silver powder.
9. Waterproof concrete should be continuously poured, and construction joints should be left less.When the construction joint is left, the shear and bending moment should not be left at the maximum or between the bottom plate and the side plate,it should be left at the top of the floor surface of not less than 300mm side wall.When the side wall has reserved holes, the edge of the construction hole should not be less than 300mm.
10. The thickness of the protective layer of the reinforcement of different parts:
The top and bottom of the roof were 35mm; The top of the floor are 50mm,The bottom of the floor are 75mm ; The reinforced concrete beam and column are 50mm;The pool walls are 50mm.
11. The soil in the 1m range around the pool wall should permeate well.The soil should be symmetrically layered and backfilled, the layer thickness is not greater than 300mm, and the compaction coefficient is not less than 0.94.
12. The building construction acceptance should comply with the current construction acceptance specification.
13. The construction of the project should be combined with the construction, water, warm, electricity and other types of drawings matching construction.
14. Other matters should be required to be strictly in accordance with the relevant procedures, specifications, standards and other related contents of the atlas implementation.
15. The initial design of the project has not yet been approved by NTDC, The possibility of revision still exists. Please contact the owner and designer before construction.
16. The reinforcement at the pipe hole shall be bypassed as far as possible and should not be cut off.When the opening diameter is not greater than 300mm, the stress steel bar can be bypassed.When the opening diameter is greater than 300mm, and there is no concentrated load around the entrance, reinforcement reinforcement should be installed at each side of the hole.
17. The anchorage length of longitudinal tensile reinforcement:

Type of bar	Class of seismic measure	Diameter of bar	Concrete strength fc' 21.0	28.0
		10	53d	52d
		13	53d	45d
		16	53d	45d
		19	53d	45d
		22	52d	45d
		25	52d	45d

Notes:
1. In any case, the anchorage length of the longitudinal tensile bar shall not be less than 250mm .
2. When the end of longitudinal tensile bar is bent or mechanically anchored, The anchor length (projection length) including the hook or anchor end,is 0.6 times the figure in the table.
3. The minimum anchorage length of longitudinally compression bar shall not be less than 0.7 times the figures in the table.

18. When the longitudinal compressive bar is overlapped, the compressive lap length shall not be less than 0.7 times of the length of the longitudinal tensile bar, and shall not be less than 200mm in any case.

The lashing lap length of longitudinal tensile bars L_l　　$L_l=L_aX$ 1.2 (≤25%) / 1.4 (50%) / 1.6 (100%)
The numbers in brackets are the percentage of overlap area of the bars in the same connecting section



孔洞处加固图
Reinforcement details at hole

施工缝节点
Construction joints

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

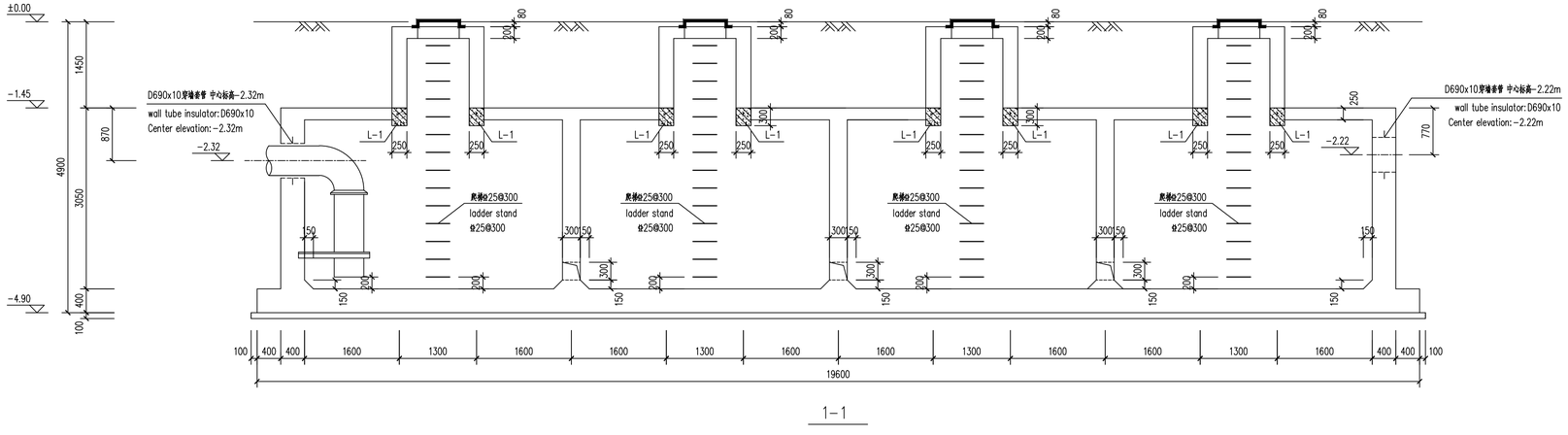
M L H V D C - L C S - E C 1 - D D - C S - 0 5 0 3 - 0 0 1 B

	姓名 Name	日期 Date
设计 Designed	刘康平	2019.06.28
校核 Checked	鲍伟	
审核 Reviewed	吕雪强	
批准 Approved	杨飞	

地下构筑物结构设计说明
Specification of structural design for underground structures

比例 Scale: 1:100　图号 DWG NO.: 30-B003701S-T0503-001　版本 REV.: B

专业 Speciality	会签 Sign	日期 Date
水工	罗鸣	



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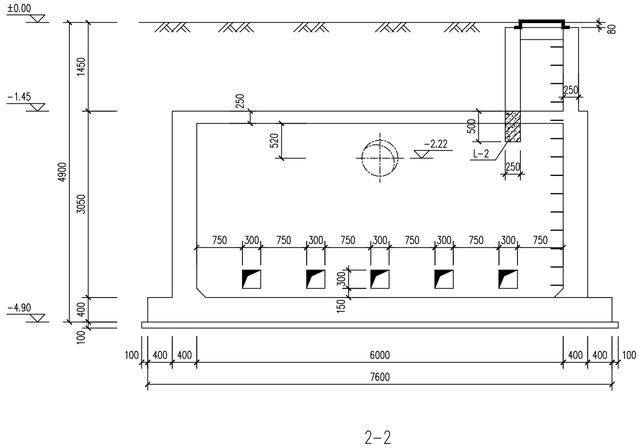

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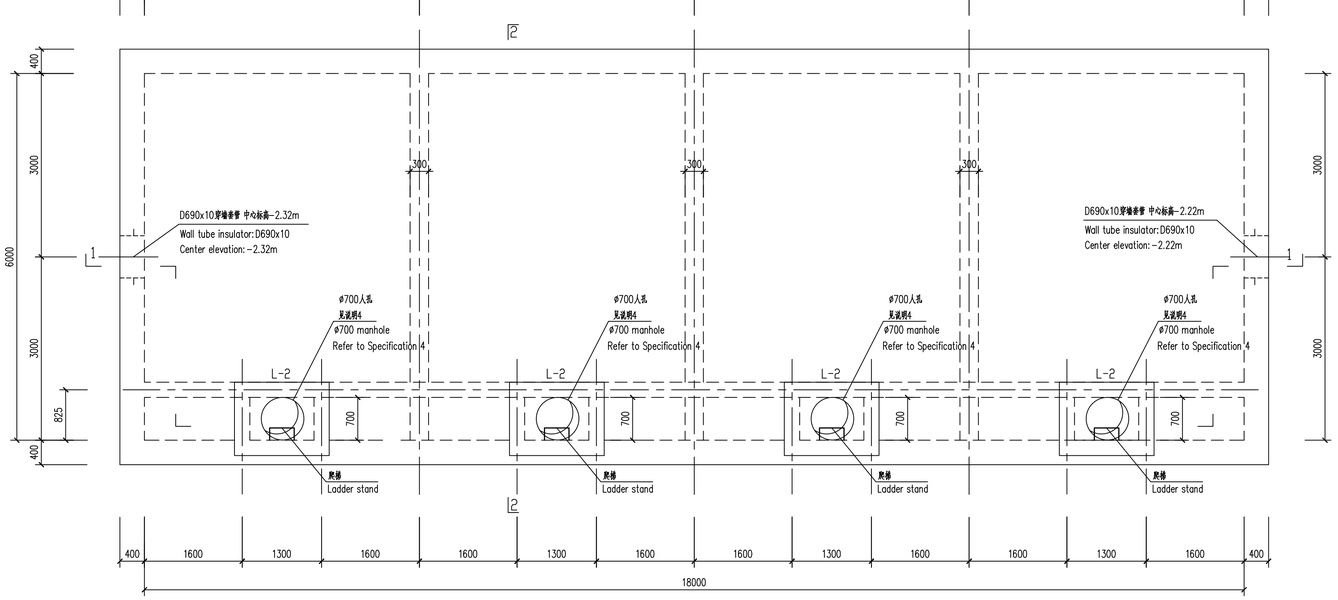
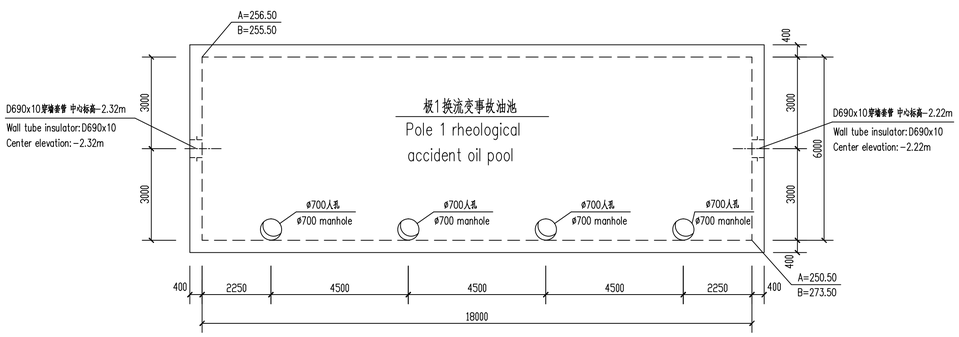


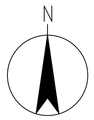
极1换流变事故油池平面图 1:50

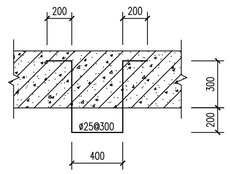
Plan of Pole 1 rheological accident oil pool



极1换流变事故油池布置图

Layout of Pole 1 rheological accident oil pool





爬梯详图

Ladder detail

说明：

1. 场地标高为±0.00m，相当于绝对标高193.40m（EGM2008高程系统）。
2. 事故油池有效容量175m³。
3. 安装管子时参见排油系统图，插口接头处需沥青嵌缝，水泥砂浆封口。
4. 放置φ700复合高分子材料盖板，详见给水排水标准图集15S501-3。
5. 外露铁件热镀锌防腐。
6. 套管为A型刚性，详见国标02S404。套管规格与进或出水管尺寸相对应。
7. 池壁四周填土应采用透水性好的土壤，基坑回填土的压实系数不小于0.94。
8. 换流变事故油池出水管需弯入池内，管子端部距油池底板200mm，
   支架和管卡详图参见给水排水标准图集S161:55-50和55-17。
9. 基础持力层位于①粉质黏土上，fak=70kPa。

Specification:

1. The site level of ±0.000 is equal to the absolute elevation of 193.40 (geodestic datum system EGM2008).
2. Effective capacity of accident oil pool: 117m³.
3. Refer to the drain system diagram for pipe installation.Socket joints require asphalt caulking, cement mortar seal
4. Place φ700 composite polymer cover.Refer to Water Supply and Drainage Standard Atlas-15S501-3.
5. All exposed iron parts are hot-dip galvanized anti-corrosion.
6. Casing type: A rigid; Refer to National standard 02S404.The casing size corresponds to the size of the inlet or outlet
7. Pond filling around the pond should be used in a good soil permeability. The backfilling of foundation pit compaction factor is not less than 0.94.
8. Accident oil pool outlet pipe to be bent into the pool. Pipe ends 200mm from the tank floor. Refer to the water supply and drainage standard photos for details of bracket and pipe clamps
9. The base holding layer is located on silt clay①, fak=70kPa.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 5 0 3 - 0 0 2 B

	姓名 Name	日期 Date
设计 Designed	刘康宁	2019.06.28
校核 Checked	鸿伟	
审核 Reviewed	张雷	
批准 Approved	杨工	

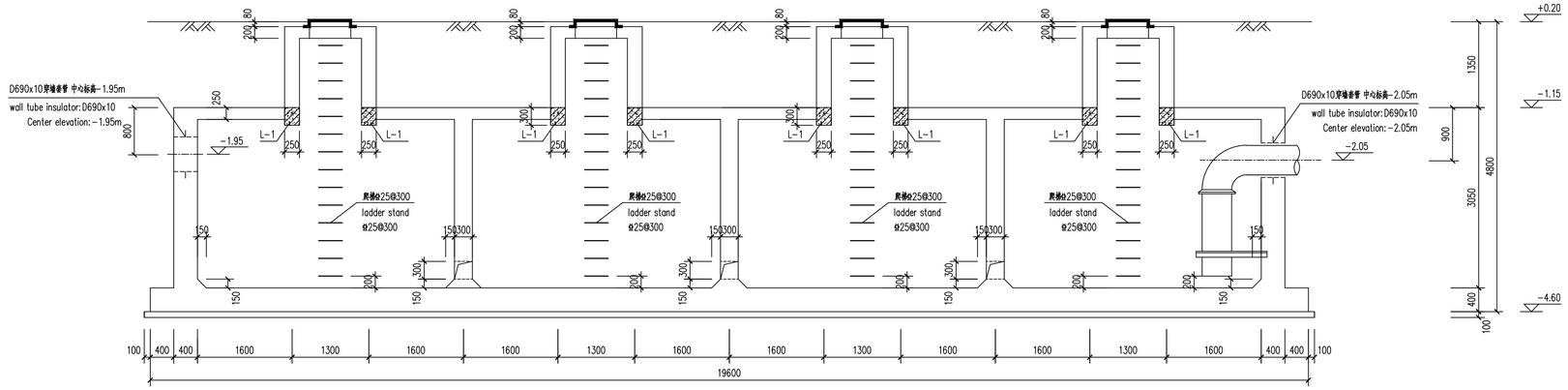
极1事故油池布置图

Drawings of Pole 1 accident oil pool

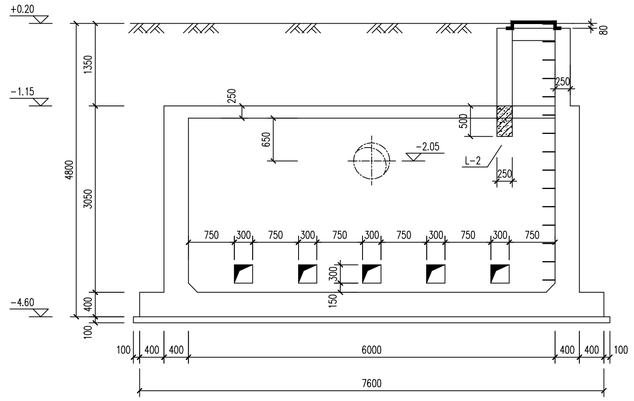
比例 Scale	图号 DWG NO.	版本 REV.
1:50	30-B003701S-T0503-002	B

专业 Speciality	会签 Sign	日期 Date
水工	甲鸿	

594X841



1-1



2-2

说明:

1. 场地标高为±0.00m，相当于绝对标高193.40m（EGM2008高程系统）。
2. 事故油池有效容量175m³。
3. 安装管子时参见排油系统图,插口接头处需沥青嵌缝,水泥砂浆封口。
4. 放置ø700复合高分子材料盖板,详见给水排水标准图集15S501-3。
5. 外露铁件热镀锌防腐。
6. 套管为A型刚性，详见国标02S404。套管规格与进或出水管尺寸相对应。
7. 池壁四周填土应采用透水性好的土壤，基坑回填土的压实系数不小于0.94。
8. 换流变事故油池出水管需弯入池内，管子端部距油池底板200mm，
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9. 基础持力层位于①粉质黏土上，fak=70kPa。

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3. Refer to the drain system diagram for pipe installation.Socket joints require asphalt caulking, cement mortar seal
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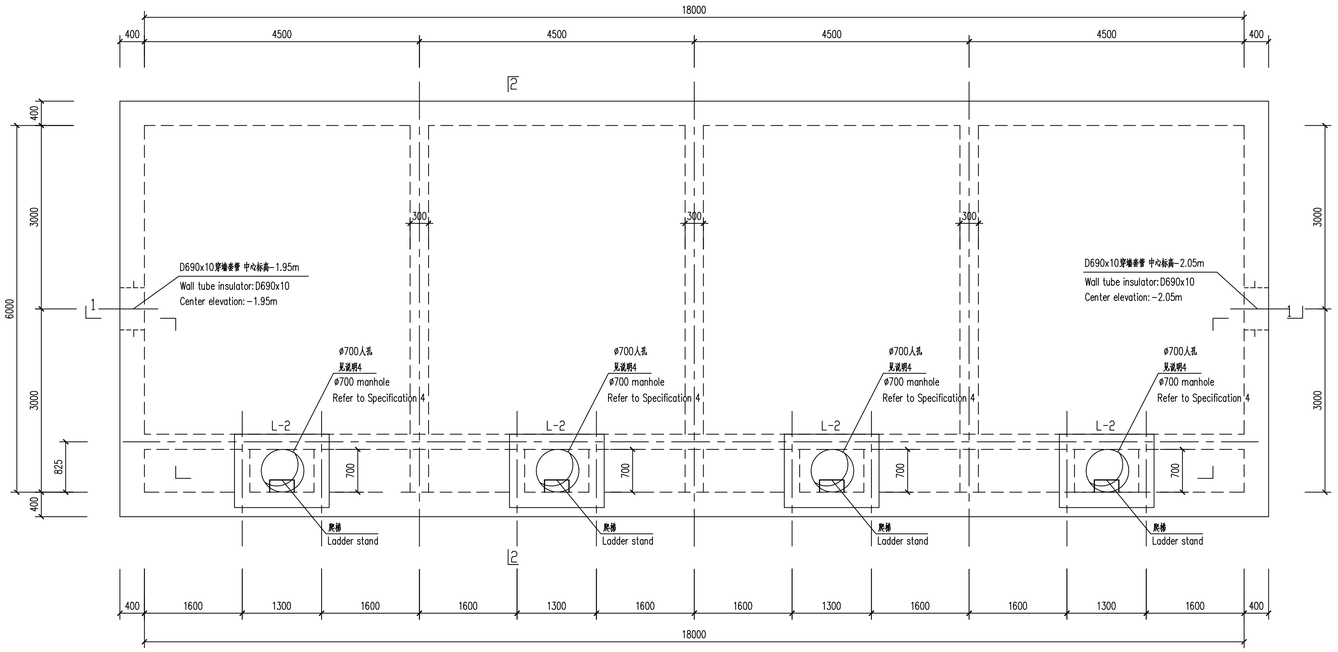
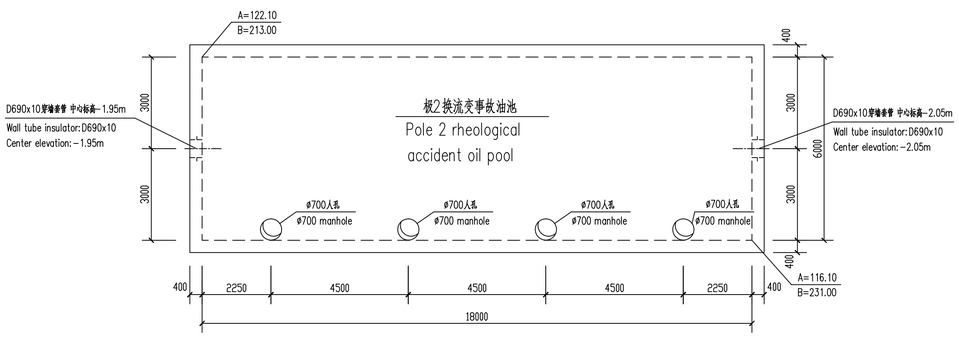
6. Casing type: A rigid; Refer to National standard 02S404.The casing size corresponds to the size of the inlet or outlet
7. Pond filling around the pond should be used in a good soil permeability. The backfilling of foundation pit compaction factor is not less than 0.94.
8. Accident oil pool outlet pipe to be bent into the pool. Pipe ends 200mm from the tank floor. Refer to the water supply and drainage standard photos for details of bracket and pipe clamps
9. The base holding layer is located on silt clay①, fak=70kPa.



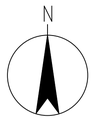
极2换流变事故油池平面图 1:50

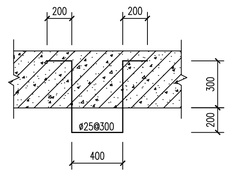
Plan of Pole 2 rheological accident oil pool



极2换流变事故油池布置图

Layout of Pole 2 rheological accident oil pool





爬梯详图

Ladder detail

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 5 0 3 - 0 0 3 B

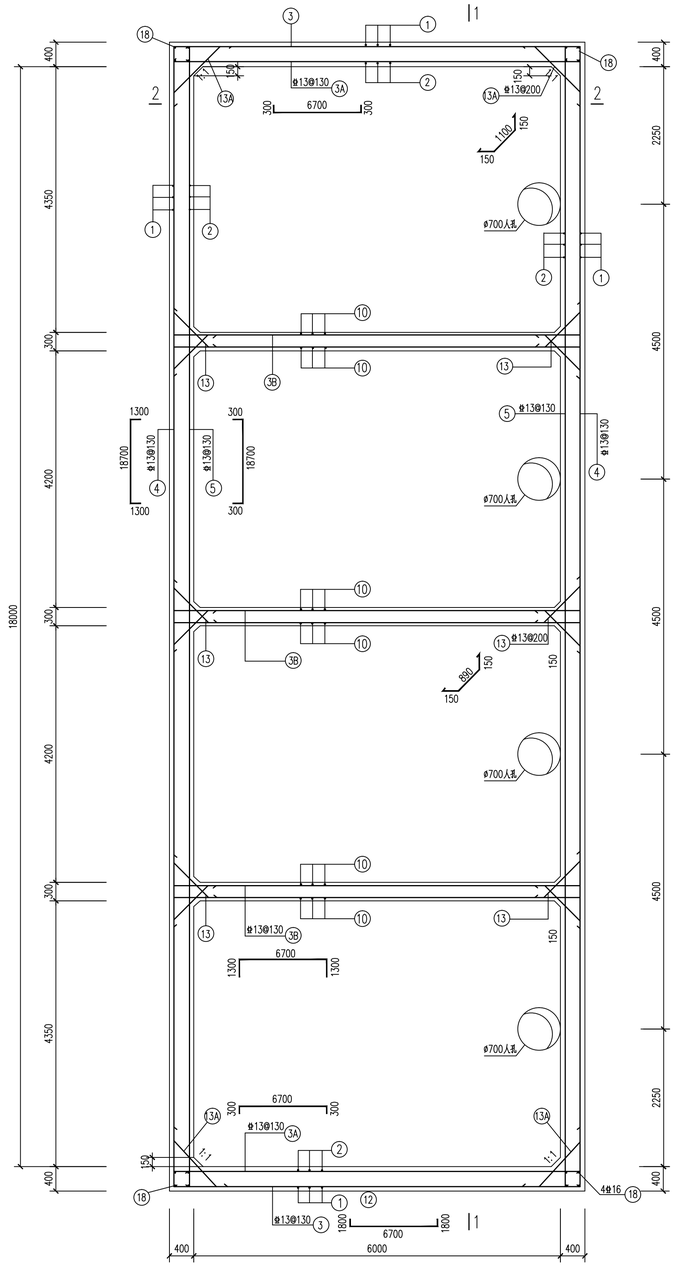
	姓名 Name	日期 Date
设计 Designed	刘康宁	2019.06.28
校核 Checked	鸿伟	
审核 Reviewed	彭雷霆	
批准 Approved	杨志	

极2事故油池布置图

Drawings of Pole 2 accident oil pool

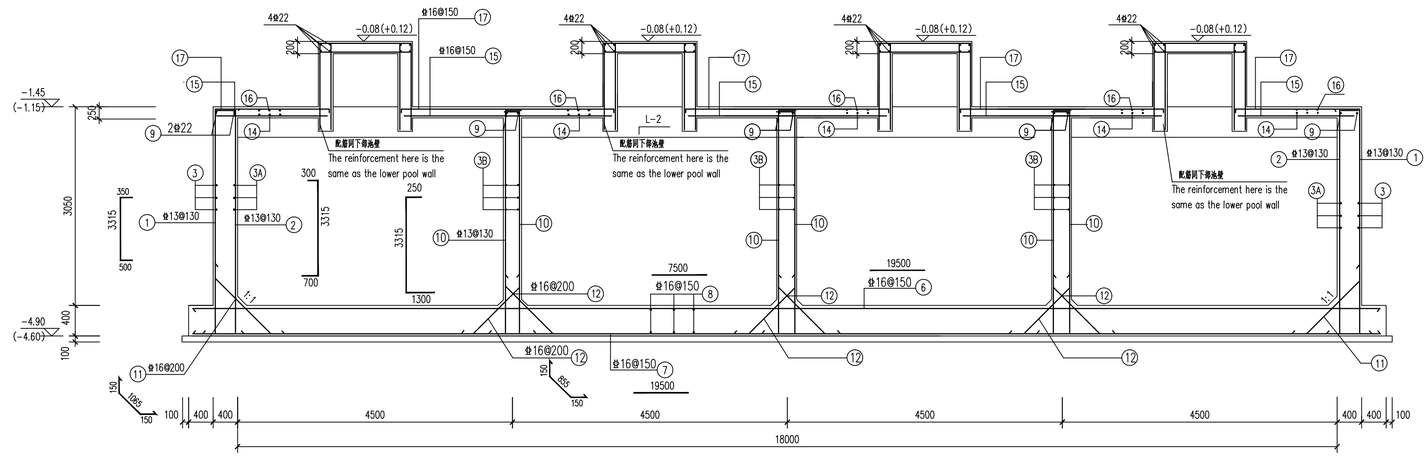
比例 Scale: 1:50 | 图号 DWG NO.: 30-B003701S-T0503-003 | 版本 REV.: B

专业 Speciality	会签 Sign	日期 Date
水工	甲	

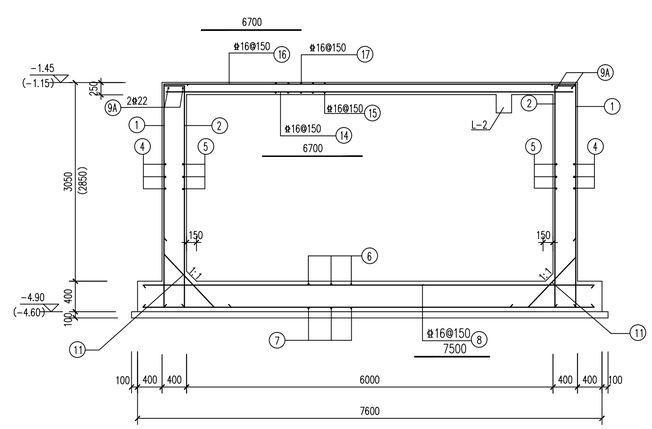


事故油池配筋平面图 1:50

Reinforcement plan of accident oil pool



1-1 1:50



2-2 1:50

说明:

1.套管,连通孔,进出水管位置详见02、03图。

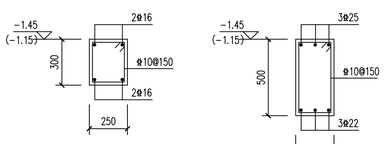
2.极2事故油池配筋同极1事故油池,括号内标注为极2事故油池。

Specification:

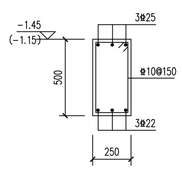
1.The position of Cannula, the communication hole,inlet and outlet pipe refer to Drwg.2,Drwg.3.

2. The reinforcement of Pole 2 accident pool is the same with Pole 1 accident pool.

Marks in parentheses is used in the Pole 2 accident oil pool.



L-1



L-2

水工	罗瑞	
专业 Speciality	会签 Sign	日期 Date

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

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M L H V D C - L C S - E C I - D D - C S - 0 5 0 3 - 0 0 4 B

	姓名 Name	日期 Date
设计 Designed	刘康平	2019.06.28
校核 Checked	鸿伟	
审核 Reviewed	苏德清	
批准 Approved	杨立	
比例 Scale	1:50	图号 DWG NO. 30-B003701S-T0503-004

事故油池配筋图

Reinforcement of accident oil pool

版本 REV. B

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